Data-Driven Customer Engagement Platform — Solution Design

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1. Glossary

Term	Definition
Airflow	Apache Airflow — open-source workflow orchestration platform for scheduling and monitoring data pipelines
APNs/FCM	Apple Push Notification service / Firebase Cloud Messaging — mobile push notification delivery services for iOS and Android
CAPEX	Capital Expenditure — one-time development and setup costs
DAG	Directed Acyclic Graph — the unit of work in Airflow; defines the sequence and dependencies of pipeline tasks
dbt	Data Build Tool — SQL-first transformation framework for analytics engineering
ETL	Extract, Transform, Load — pattern for moving data from source systems into an analytical store
GDPR	General Data Protection Regulation — EU data privacy regulation requiring explicit consent for personal data use
KPI	Key Performance Indicator — measurable value showing how effectively business objectives are being achieved
MoSCoW	Prioritisation framework: Must have, Should have, Could have, Won’t have
MWAA	Amazon Managed Workflows for Apache Airflow — fully managed Airflow service on AWS; removes operational burden of self-hosting
OLAP	Online Analytical Processing — database optimised for queries and analysis (e.g. Snowflake)
OLTP	Online Transactional Processing — database optimised for day-to-day transactions (e.g. WordPress MySQL)
OPEX	Operational Expenditure — ongoing monthly costs (infrastructure, licences, tooling)
SES	Simple Email Service — AWS managed email sending service
Snowflake	Cloud-based OLAP data warehouse — managed service on AWS, SQL-first, scales compute independently of storage
TCO	Total Cost of Ownership — full cost including development, infrastructure, and team over a given period
VPN	Virtual Private Network — encrypted tunnel between on-premises infrastructure and cloud

2. Executive Summary

2.1. Context

Your organisation is an established European grocery and commodity webshop — 15 years on the market, ~1M daily visitors, 2M mobile app installs. The goal is to increase annual turnover by 20%. Today, customer data is scattered across isolated systems: the WordPress e-commerce platform, the mobile application, and wholesaler product feeds each operate independently, with no unified view of purchasing behaviour.

This document proposes a unified data pipeline that consolidates customer, product, and wholesaler data into a single analytical layer — and uses it to deliver personalised product suggestions via email and mobile push notifications. The platform is designed to be owned and operated by your existing team, without requiring a dedicated data function.

2.2. Recommendation at a Glance

Aspect	Summary
Recommended Option	Option 2 — Snowflake Data Platform with Managed ETL
Estimated CAPEX	~$500K (within confirmed budget; incl. 17% risk buffer)
Estimated OPEX	$6K–$16K/month (Snowflake + AWS SES + MWAA + infrastructure)
Timeline	7 months (phased delivery)
Team Size	5 engineers: 2 Data Engineers + 1 Backend + 1 ML/Data Scientist + 1 QA + 0.5 SA
Go-Live Target	April 2027 (next fiscal year), with proved results before 2028

2.3. Scope Boundary

This engagement delivers: unified data pipeline (WordPress, mobile app, wholesalers), Snowflake analytical platform, customer segmentation and recommendation engine, email campaign delivery via AWS SES, mobile push notifications, and a self-service analytics dashboard.

This engagement explicitly **does not** include: WordPress migration or re-platforming, new feature development for the mobile application, EDI or real-time API integration with wholesalers, real-time personalisation, or recruitment of a permanent data team. Any change to this scope is subject to a formal change request and re-estimation.

3. Problem Statement

3.1. Current Pain Points

#	Problem	Business Impact
1	Customer data scattered across isolated systems (WordPress DB, mobile app, wholesaler feeds) — no unified view	Cannot identify cross-channel buying patterns; personalisation impossible
2	No mechanism to proactively suggest products to existing customers	~1M daily visitors but no re-engagement; revenue limited to organic visits
3	Mobile app data accessible via vendor APIs, but 2M installs = primary channel data missing from analytics	Mobile customer behaviour invisible; cannot target 2M app users with personalised offers
4	15 years of accumulated customer emails — many stale or invalid (10-minute mailboxes)	Email marketing ROI degraded by poor deliverability
5	Internal team lacks data engineering expertise to build analytics capabilities	Data-driven decisions delayed; business relies on intuition over evidence

3.2. Target State

The enterprise has a unified data platform that continuously consolidates customer, product, and wholesaler data from all sources into a single analytical store. A personalised suggestion engine analyses buying patterns and matches customers with relevant products and deals, delivering timely recommendations via email and mobile push. The marketing team monitors campaign performance through self-service dashboards, and the platform is designed for the internal team to gradually take ownership with minimal external dependency.

3.3. Business Goal Translation

Your stated goal is "increase turnover by 20%." In technical terms this means:

- **Unify** all customer data islands into a single source of truth
- **Profile** customers based on purchase history, browsing behaviour, and preferences
- **Recommend** products with the highest conversion probability per customer segment
- **Deliver** recommendations through email (AWS SES) and mobile push (2M installs)
- **Measure** uplift via campaign tracking and A/B testing in the analytics dashboard

4. Solution Options

4.1. Side-by-Side Comparison

Aspect	Option 1: Quick Win	Option 2: Snowflake Platform ✅	Option 3: Databricks Full AI
Scope	WordPress data only + basic email suggestions via SQL segmentation	All data sources + Snowflake + ML-based suggestions + email & mobile	All sources + Databricks + custom AI model training + real-time streaming
Data Sources	WordPress DB only	WordPress + Mobile (API) + Wholesalers	All sources + real-time event streams
AI Capability	Rule-based (SQL queries)	ML-based segmentation (Snowflake ML or external)	Custom-trained recommendation models (Databricks MLflow)
Team	2 devs (1 Data Eng + 1 Backend)	5 devs (2 Data Eng + 1 Backend + 1 ML + 1 QA)	8+ devs (3 Data Eng + 2 ML + 1 Backend + 1 DevOps + 1 QA)
Timeline	3 months	7 months (phased)	12+ months
CAPEX (incl. risk buffer)	$80K–$150K	~$500K (within budget)	$800K–$1.5M
OPEX / month	$2K–$5K	$6K–$15K	$25K–$50K
Risk	Low (limited scope, fast delivery)	Medium (vendor negotiation, data quality)	High (complexity, team skill gap, long timeline)

Option 1 scores better on cost and operational risk — a valid choice for an organisation prioritising simplicity over growth. Option 2 is recommended because your 20% turnover target requires ML-based segmentation, multi-source data consolidation, and multi-channel delivery that Option 1 cannot provide. Option 3 was not taken forward — at $800K–$1.5M it exceeds the confirmed $500K budget, and the additional AI capability it offers is not required for the current scope.

5. Recommended Option: Snowflake Data Platform with Managed ETL

5.1. System Context

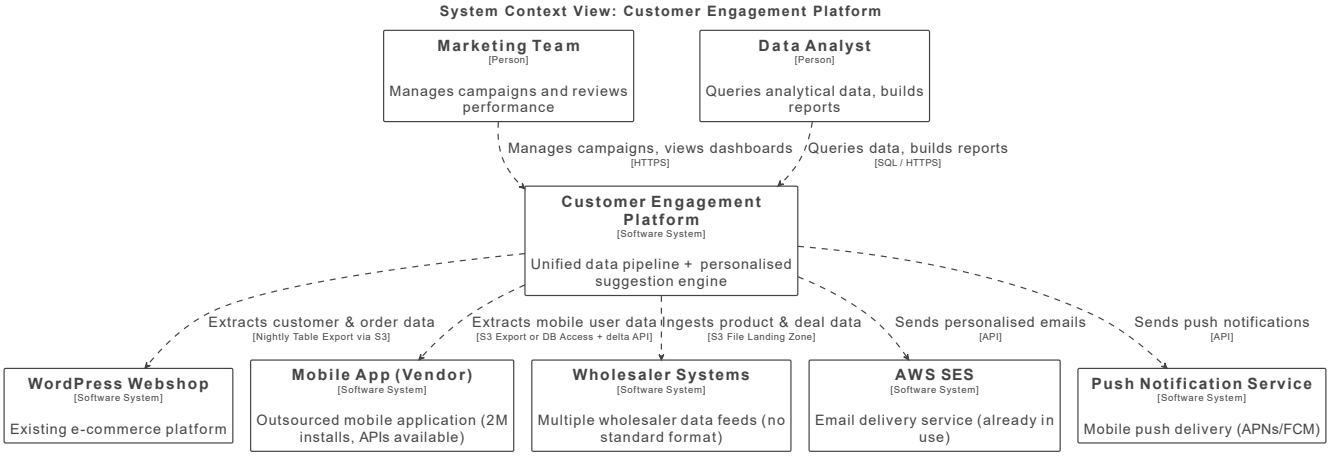

Figure 1. System Context — Data-Driven Customer Engagement Platform

5.2. High-Level Architecture

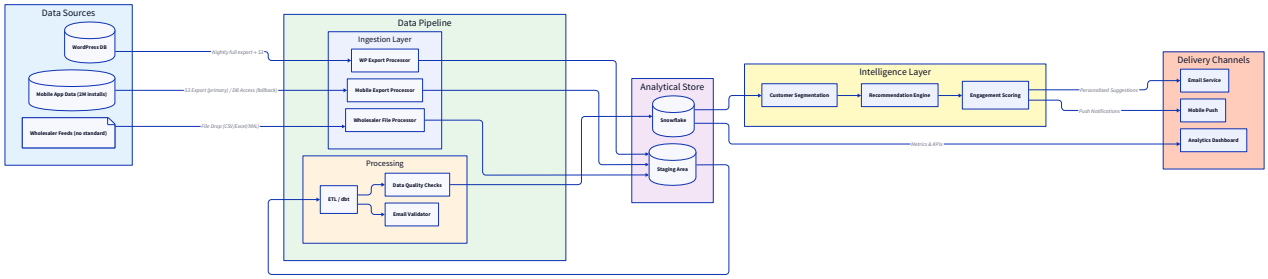

Figure 2. Architecture — Option 2: Snowflake Data Platform

5.3. What It Delivers

- Unified analytical data store (Snowflake) consolidating WordPress, mobile, and wholesaler data
- Automated ETL pipeline with data quality validation and email address verification
- ML-based customer segmentation and personalised product recommendation engine
- Personalised email campaign delivery via **AWS SES** (already in use)
- Mobile push notification capability (Phase 2 — 2M installs makes this a **primary channel**)
- Self-service analytics dashboard for marketing team (campaign performance, KPIs)
- GDPR-compliant data processing with consent management and customer deletion support

5.4. Pros & Cons

Pros	Cons
Snowflake is mature, well-documented, popular on AWS — strong community and hiring pool	Snowflake licence cost scales with usage — unpredictable at high query volumes
Managed service reduces operational burden — fits non-technical internal team	Vendor lock-in risk — Snowflake is proprietary
Phased delivery reduces risk — email channel deliverable within 3-4 months	Mobile vendor cooperation is uncertain — Phase 2 may be delayed
Covers all three data sources — provides the unified view your organisation needs	Data quality of 15-year-old records is unknown — cleaning effort may be substantial
Extensible to AI/ML workloads in future (Snowpark, external ML tools)	Your team will need training to operate the platform: Snowflake (cost monitoring, queries), Airflow (pipeline status), Metabase (dashboards), email campaign workflow, and the GDPR deletion process. Training is included in scope.

6. Functional Requirements